

I&C Fundamentals

Drill Input Library

Reusable bracket banks for active recall, loop reconstruction, calibration, and troubleshooting practice.

01

Pick a bracket

Choose [fundamental], [device], [loop], [symptom], or another input.

02

Run the drill

Answer from memory before checking references.

03

Correct + repeat

Rewrite misses until the signal path or concept is clean.

Core progression

Electrical -> Signals -> Measurement -> Control -> Drawings -> Calibration -> Troubleshooting

HOW TO USE THIS PDF Do not try to memorize the entire library. Pick one input, retrieve it cold, apply it to a real loop or symptom, correct the miss, then revisit it later.

Quick Navigation

Use this page when you know the kind of drill you need but not the exact input.

Situation	Bracket	Use it for
Need raw fundamentals?	[fundamental]	Ohm's law, 4-20 mA, RTDs, DP, valves, PID, PLC I/O
Need the why?	[concept]	Physical mechanisms and cause-and-effect relationships
Mixing two ideas up?	[concept A] vs [concept B]	Discrimination drills: sourcing/sinking, LRV/URV, RTD/TC
Need hardware practice?	[device]	Transmitters, switches, valves, I/Ps, PLC modules, calibrators
Need signal-path practice?	[loop]	Measurement, discrete, and control loops
Need fault-finding?	[symptom]	Realistic failures that force evidence-based troubleshooting
Need print literacy?	[drawing]	P&ID, loop sheet, ladder, termination, I/O, cause/effect
Need scaling practice?	[LRV] / [URV]	Ranges that prevent memorized 0-to-full-scale shortcuts

Priority rule: spend more repetitions on troubleshooting and complete signal paths than on isolated vocabulary.

I'd use the banks below as the **actual inputs for the drill pack**.

The ordering is intentional. ISA's current CCST framework is based on technician job-task analysis, and Level 1 heavily emphasizes calibration, maintenance, repair, troubleshooting, loop checking, commissioning, and documentation. ISA also explicitly centers pressure, temperature, level, flow, P&IDs, and calibration principles in its technician curriculum.

That matches experienced-tech feedback surprisingly well: strong technicians are repeatedly described as people who understand the underlying measurement principle, can read the drawings, understand the process, and **fault-find logically rather than swap parts**.

[fundamental]

This should be your **largest input bank**. Use one item at a time.

Electrical fundamentals

- Ohm's law
- Voltage
- Current
- Resistance
- Electrical power
- Voltage drop
- Series circuits
- Parallel circuits
- Series-parallel circuits
- Open circuits
- Short circuits
- Continuity
- DC polarity
- 24 VDC control power
- Fuse protection
- Circuit common
- Ground vs common
- Normally open contacts
- Normally closed contacts
- Sourcing inputs
- Sinking inputs
- PNP devices
- NPN devices

Analog signal fundamentals

- 4-20 mA current loop
- Live zero
- Loop power
- Loop resistance
- Loop burden
- Compliance voltage
- 0-10 V signals
- Current vs voltage signals
- 2-wire transmitters
- 3-wire transmitters
- 4-wire transmitters
- Active analog signals
- Passive analog signals
- Analog signal scaling
- Percent-of-span conversion
- HART fundamentals
- HART communication over 4-20 mA

A standard current loop fundamentally consists of power, transmitter, receiver/controller, and wiring, with Ohm's law underlying loop voltage, resistance, and current behavior.

Transmitter fundamentals

- Sensor vs transmitter
- Process variable
- LRV
- URV
- Range
- Span
- Zero
- Zero suppression
- Zero elevation
- Direct acting transmitter
- Reverse acting transmitter
- Engineering-unit scaling
- Transmitter accuracy
- Calibration tolerance
- Linearity
- Repeatability
- Hysteresis
- Sensor error
- Transmitter error
- As-found condition
- As-left condition

Pressure

- Gauge pressure
- Absolute pressure
- Differential pressure
- Atmospheric pressure
- Static pressure
- Differential-pressure transmitter
- Pressure transmitter
- Impulse lines
- High-pressure side
- Low-pressure side
- Wet leg
- Dry leg
- Pressure measurement using diaphragms

Level

- Hydrostatic level measurement
- Differential-pressure level
- Open-tank DP level
- Closed-tank DP level
- Wet-leg level measurement
- Dry-leg level measurement
- Specific gravity effects
- Density effects on DP level
- Float level measurement
- Displacer fundamentals
- Ultrasonic level fundamentals
- Radar level fundamentals
- Guided-wave radar fundamentals

Flow

- Differential-pressure flow

- Orifice plates
- Primary flow elements
- Square-root relationship
- Volumetric flow
- Mass flow
- Magnetic flowmeter principle
- Vortex flowmeter principle
- Coriolis principle
- Turbine flowmeter principle
- Straight-run effects
- Flow direction

Temperature

- RTD principle
- Resistance vs temperature
- 2-wire RTD
- 3-wire RTD
- 4-wire RTD
- Lead-wire resistance
- Thermocouple principle
- Seebeck effect
- Thermocouple polarity
- Cold-junction compensation
- Thermowells
- Temperature transmitters
- Sensor vs process temperature
- Sensor response time

Control valves

- Control valve purpose
- Valve actuator
- Pneumatic actuator
- Spring-and-diaphragm actuator
- I/P converter
- Valve positioner
- Valve position feedback
- Air-to-open
- Air-to-close
- Fail-open
- Fail-closed
- Fail-in-place
- Direct valve action
- Reverse valve action
- Cv
- Valve travel
- Valve stiction
- Valve deadband
- Instrument air
- 3-15 psi pneumatic signal
- 4-20 mA to pneumatic conversion

Basic process control

- Process variable
- Setpoint
- Manipulated variable
- Control output
- Error
- Open-loop control

- Closed-loop control
- Feedback
- Negative feedback
- Controller action
- Direct acting controller
- Reverse acting controller
- Proportional control
- Proportional gain
- Proportional band
- Integral control
- Reset
- Derivative control
- Rate
- PID control
- Dead time
- Process lag
- Oscillation
- Offset
- Integral windup
- Cascade control
- Feedforward control

PLC/control-system fundamentals

- Digital input
- Digital output
- Analog input
- Analog output
- PLC input card
- PLC output card
- Field input
- Field output
- Input image/table fundamentals
- Output image/table fundamentals
- Ladder logic
- Examine-if-closed concept
- Examine-if-open concept
- Output coil
- Seal-in circuit
- Start-stop logic
- Permissive
- Interlock
- Alarm
- Trip
- PLC scan cycle
- Analog scaling in PLC logic

Documentation

- P&ID
- Process flow diagram
- Instrument loop diagram
- Electrical schematic
- Wiring diagram
- Termination drawing
- I/O list
- Instrument index
- Instrument datasheet
- Cause-and-effect diagram
- Ladder diagram

- Panel drawing
- Cable schedule
- Instrument tag identification
- ISA instrument symbols
- Signal-line symbols

ISA-5.1 specifically standardizes instrumentation symbols and identification, while ISA-5.4 covers loop diagrams and their use in installation, operation, troubleshooting, and maintenance.

Installation and signal integrity

- Grounding
- Shielding
- Single-point shield grounding
- Ground loops
- Electrical noise
- EMI
- RFI
- Twisted-pair wiring
- Cable segregation
- Signal isolation
- Instrument grounding
- Intrinsic safety
- Intrinsically safe circuit
- Intrinsic safety barrier
- Galvanic isolator
- Hazardous area classification
- Class/Division fundamentals
- Zone fundamentals

Calibration and commissioning

- Calibration
- Verification
- Functional test
- Loop check
- Point-to-point check
- Five-point calibration
- Zero check
- Span check
- Rising calibration
- Falling calibration
- Calibration tolerance
- Error calculation
- Percent-of-span error
- Simulating mA
- Sourcing mA
- Measuring mA
- Loop power
- Calibration standard
- Traceability
- As-found readings
- As-left readings
- Loop-check documentation

[concept]

Use this when the drill specifically asks you to **understand why something works** rather than merely recall facts.

Best inputs:

- Why current is the same throughout a series 4-20 mA loop

- Why 4 mA is used instead of 0 mA
- How voltage drops distribute around a current loop
- Why excessive loop resistance causes problems
- How a 2-wire transmitter powers itself
- Active vs passive current signals
- Sourcing vs sinking
- NO vs NC as physical state versus PLC logic state
- Why fail-safe contacts are often normally closed
- How a transmitter converts a process variable into 4-20 mA
- LRV vs URV
- Range vs span
- Zero shift vs span error
- Accuracy vs precision
- Calibration vs verification
- As-found vs as-left
- How an RTD measures temperature
- How a thermocouple produces a signal
- Why thermocouples require cold-junction compensation
- Why 3-wire RTDs compensate for lead resistance
- Why thermowell installation affects response
- How pressure creates force on a sensing element
- How hydrostatic pressure indicates level
- Why density affects DP level
- How differential pressure represents flow
- Why DP flow has a square-root relationship
- Sensor measurement vs inferred process measurement
- How an I/P works
- How a valve positioner closes its own position loop
- Air-to-open vs fail-closed
- Air-to-close vs fail-open
- Cv and valve capacity
- Feedback control
- Process variable vs setpoint
- Controller output vs final valve position
- Direct vs reverse controller action
- What proportional action does
- What integral action does
- What derivative action does
- Why excessive gain causes oscillation
- Why excessive integral action can cause cycling
- Dead time vs lag
- Open vs closed loop
- Cascade control
- Feedforward control
- Permissive vs interlock
- Analog input scaling
- PLC scan sequence
- Ground vs common
- Shielding vs grounding
- Why ground loops create measurement errors
- Intrinsic safety principle

[concept A] vs [concept B]

This is one of the highest-value drills because good troubleshooting depends heavily on **distinguishing things that initially look alike**.

[concept A][concept B]	
Voltage	Current
Series circuit	Parallel circuit
Open circuit	Short circuit
Sourcing	Sinking
PNP	NPN
Normally open	Normally closed
Ground	DC common
Current signal	Voltage signal
Active signal	Passive signal
2-wire transmitter	4-wire transmitter
Measuring mA	Sourcing mA
Sourcing mA	Simulating mA
LRV	URV
Range	Span
Zero error	Span error
Accuracy	Precision
Repeatability	Accuracy
Calibration	Verification
Calibration	Loop check
As-found	As-left
RTD	Thermocouple
2-wire RTD	3-wire RTD
3-wire RTD	4-wire RTD
Sensor	Transmitter
Gauge pressure	Absolute pressure
Gauge pressure	Differential pressure
Wet leg	Dry leg
Pressure	Differential pressure
Mass flow	Volumetric flow
Radar	Ultrasonic level

[concept A][concept B]	
Air-to-open	Air-to-close
Fail-open	Fail-closed
I/P	Positioner
Valve command	Valve position
Direct acting	Reverse acting
Open-loop control	Closed-loop control
Proportional	Integral
Integral	Derivative
Dead time	Process lag
Cascade	Feedforward
Process variable	Setpoint
Permissive	Interlock
Alarm	Trip
Digital input	Digital output
Analog input	Analog output
P&ID	Loop drawing
P&ID	Electrical schematic
Grounding	Shielding
Intrinsic safety barrier	Galvanic isolator

[device]

These are the real devices I would rotate through most often.

Basic electrical/control

- 24 VDC power supply
- Fuse
- Circuit breaker
- Terminal block
- Relay
- Interposing relay
- Control relay
- Selector switch
- Pushbutton
- Limit switch
- Proximity switch
- Inductive proximity sensor
- Photoelectric sensor

- Float switch
- Pressure switch
- Temperature switch
- Solenoid valve

Transmitters and sensors

- Pressure transmitter
- Differential-pressure transmitter
- Level transmitter
- Flow transmitter
- Temperature transmitter
- RTD
- Thermocouple
- Thermowell
- Magnetic flowmeter
- Coriolis flowmeter
- Vortex flowmeter
- DP flow transmitter
- Radar level transmitter
- Guided-wave radar transmitter
- Ultrasonic level transmitter

Final control

- Control valve
- Pneumatic control valve
- Valve actuator
- I/P converter
- Valve positioner
- Solenoid-operated valve
- Position feedback transmitter

Control system

- PLC
- Remote I/O rack
- Digital input module
- Digital output module
- Analog input module
- Analog output module
- HMI
- PID controller
- Signal isolator
- intrinsic-safety barrier

Test equipment

- Digital multimeter
- Loop calibrator
- Process calibrator
- Pressure calibrator
- Pressure source
- Temperature calibrator
- RTD simulator
- Thermocouple simulator

[loop]

This is probably your **most useful practical bank**.

Start with these.

Analog measurement loops

- 2-wire pressure transmitter -> PLC analog input
- 2-wire DP transmitter -> PLC analog input
- Level transmitter -> PLC analog input
- Flow transmitter -> PLC analog input
- Temperature transmitter -> PLC analog input
- RTD -> temperature transmitter -> PLC analog input
- Thermocouple -> temperature transmitter -> PLC analog input
- Magnetic flowmeter -> PLC analog input
- Radar level transmitter -> PLC analog input

Control loops

- PLC analog output -> I/P -> pneumatic control valve
- PLC analog output -> smart positioner -> control valve
- Pressure transmitter -> PID controller -> control valve
- Flow transmitter -> PID controller -> control valve
- Level transmitter -> PID controller -> control valve
- Temperature transmitter -> PID controller -> control valve

Discrete loops

- Limit switch -> PLC digital input
- Proximity sensor -> PLC digital input
- Pressure switch -> PLC digital input
- Float switch -> PLC digital input
- PLC digital output -> interposing relay -> solenoid
- PLC digital output -> solenoid valve
- PLC output -> relay -> motor starter
- Emergency/permissive contact -> PLC input

More useful combined loops

- 24 VDC supply -> transmitter -> analog input -> DC common
- Sensor -> transmitter -> PLC -> HMI
- PLC -> I/P -> actuator -> valve -> process
- Level transmitter -> PLC -> high-level alarm
- Pressure switch -> PLC -> permissive logic -> motor
- Limit switch -> PLC -> interlock -> output
- Temperature sensor -> transmitter -> PLC -> PID -> control valve

Experienced technicians repeatedly emphasize being able to follow the complete system from drawing to device and determine where the expected signal disappears. That is also reflected in ISA's troubleshooting and loop-check focus.

[tag]

You do **not** need plant-specific tag numbers for practice. Use realistic generic ones.

Tag	Practice meaning
PT-101	Pressure transmitter
PI-101	Pressure indicator
PIC-101	Pressure indicating controller
PSH-101	Pressure switch high
PSL-101	Pressure switch low
PDT-101	Differential-pressure transmitter

Tag	Practice meaning
TT-101	Temperature transmitter
TI-101	Temperature indicator
TIC-101	Temperature indicating controller
TSH-101	Temperature switch high
FT-101	Flow transmitter
FI-101	Flow indicator
FIC-101	Flow indicating controller
FSL-101	Flow switch low
LT-101	Level transmitter
LI-101	Level indicator
LIC-101	Level indicating controller
LSH-101	Level switch high
LSL-101	Level switch low
FV-101	Flow control valve
PV-101	Pressure control valve
LV-101	Level control valve
TV-101	Temperature control valve
XV-101	On/off valve
ZS-101	Position switch
ZSO-101	Open position switch
ZSC-101	Closed position switch

ISA-5.1 exists specifically to make this type of instrument identification consistent across drawings and control systems.

[symptom]

This is the bank I would spend **a lot** of time on.

It turns book knowledge into technician thinking.

4-20 mA

- Analog input reads 0 mA
- Analog input reads approximately 4 mA regardless of process
- Analog input reads approximately 20 mA regardless of process
- Signal is below 4 mA
- Signal is above 20 mA
- Transmitter display is correct but PLC value is wrong

- Transmitter output is correct but HMI value is wrong
- Field mA differs from PLC-indicated mA
- Loop works with calibrator but not transmitter
- Transmitter will not power up
- Loop voltage is too low
- Current signal fluctuates
- Signal changes when another machine starts
- Reading is noisy

Discrete/PLC

- Input LED is off when sensor is made
- Input LED is on but PLC logic does not respond
- Field switch changes but PLC input does not
- PLC output is on but field device is off
- PLC output LED is off despite command
- Solenoid is energized but valve does not move
- Motor will not start because permissive is missing
- Interlock will not clear
- Limit switch shows wrong state
- Proximity switch LED changes but PLC input does not

Pressure

- Pressure reading stuck
- Pressure reading zero
- Pressure reads higher than expected
- Pressure reads lower than expected
- DP transmitter reads backwards
- Reading changes slowly
- Pressure transmitter passes calibration but process reading is wrong
- Impulse line is plugged
- One DP impulse line is leaking

Level

- Level reads zero on a full vessel
- Level reads full on an empty vessel
- Level changes in the wrong direction
- DP level changes when vessel pressure changes
- Level reading is consistently offset
- Level transmitter responds slowly
- Radar level is unstable

Flow

- Flow reads zero while process is clearly flowing
- Flow shows when process is stopped
- Flow reading is unstable
- DP flow reads incorrectly at low flow
- Flow reading is lower than expected
- Flow reading changes after process density changes

Temperature

- RTD reads open circuit
- Temperature reads extremely high
- Temperature reads extremely low
- Temperature jumps intermittently
- Temperature transmitter reads correctly but PLC does not
- Thermocouple polarity is reversed
- Temperature responds very slowly
- RTD lead resistance causes measurement error

Valves

- Valve does not move
- Valve only partially strokes
- Valve moves opposite the command
- Valve command changes but position does not
- Valve position changes but process does not respond
- Valve oscillates
- Valve sticks
- Valve reaches 100% command but not 100% travel
- Valve fails closed unexpectedly
- Valve fails open unexpectedly
- I/P receives 4-20 mA but produces no air output
- Positioner input is correct but valve position is wrong
- Instrument air is lost

PID/process control

- Process continuously oscillates
- Process responds too slowly
- Controller output hunts
- Process never reaches setpoint
- Large overshoot after setpoint change
- Integral action keeps driving output
- Controller responds in the wrong direction
- Output reaches 100% but PV does not reach setpoint
- Process reacts long after output changes
- Loop works manually but poorly in automatic

[drawing]

Rotate these rather than practicing only P&IDs.

- P&ID
- Process flow diagram
- Instrument loop diagram
- Loop sheet
- Electrical schematic
- Wiring diagram
- Termination diagram
- Control panel drawing
- Junction-box drawing
- PLC I/O drawing
- Ladder logic diagram
- Cause-and-effect diagram
- I/O list
- Instrument index
- Cable schedule
- Instrument datasheet

For actual technician work, **P&IDs + loop diagrams + electrical/termination drawings + PLC logic** are the combination I would prioritize. ISA explicitly describes loop diagrams as useful for installation, operation, troubleshooting, and maintenance.

[receiver or PLC input]

Use:

- PLC analog input
- DCS analog input
- Remote analog input module
- Panel meter

- Digital indicator
- PID controller input
- Signal isolator input
- Data-acquisition input
- PLC digital input
- Remote digital input module

For your drills, **PLC analog input** should be the default most of the time.

[LRV] and [URV]

Don't use the same range repeatedly. That makes scaling exercises too predictable.

Measurement[LRV][URV]		
Pressure	0 psi	100 psi
Pressure	0 psi	250 psi
Pressure	-10 psi	50 psi
Pressure	20 psi	120 psi
Differential pressure	0 inH ₂ O	100 inH ₂ O
Differential pressure	-25 inH ₂ O	25 inH ₂ O
Level	0 ft	20 ft
Level	2 ft	18 ft
Flow	0 GPM	500 GPM
Flow	0 GPM	1,200 GPM
Flow	0 lb/hr	20,000 lb/hr
Temperature	0 °F	300 °F
Temperature	-40 °F	250 °F
Temperature	50 °F	450 °F
Temperature	0 °C	200 °C

Especially practice ranges where **LRV ≠ 0**. Those expose whether you actually understand scaling rather than memorizing:

4 mA = 0

20 mA = full scale

[fundamental 1] through [fundamental 5]

For the spaced-rotation prompt, I would **interleave unrelated areas intentionally**.

Instead of:

pressure, pressure transmitter, pressure calibration, DP pressure, pressure switch

use something like:

SlotGood rotating input	
[fundamental 1]	Ohm's law
[fundamental 2]	4-20 mA scaling
[fundamental 3]	RTD principle
[fundamental 4]	control valve fail action
[fundamental 5]	PLC permissives

Then another session:

SlotInput	
1	sourcing vs sinking
2	DP level
3	thermocouple cold junction
4	proportional control
5	P&ID tag identification

Then:

SlotInput	
1	series vs parallel
2	flow measurement
3	LRV/URV/span
4	I/P operation
5	intrinsic safety

That **interleaving matters** because the brain can't simply recognize the category and coast.

[pressure / level / flow / temperature]

Use all four repeatedly, but vary the measurement principle underneath them.

VariableRotate through	
Pressure	gauge, absolute, differential, diaphragm sensing, impulse piping
Level	hydrostatic, DP, float, ultrasonic, radar, guided-wave radar
Flow	DP/orifice, magnetic, vortex, Coriolis, turbine
Temperature	RTD, thermocouple, temperature transmitter, thermowell

My Priority Bank for You

If the purpose is "make the fundamentals second nature", I wouldn't try to cycle through all 200+ inputs equally.

I would hammer these first:

Electrical -> signal -> measurement -> control -> troubleshooting

```

Ohm's law
v
Series / parallel
v
24 VDC circuits
v
NO / NC
v
Sourcing / sinking
v
4-20 mA
v
2-wire transmitters
v
LRV / URV / span / scaling
v
Pressure
v
Level
v
Flow
v
RTD
v
Thermocouple
v
Control valves
v
I/P
v
Positioner
v
PLC DI / DO
v
PLC AI / AO
v
Permissives / interlocks
v
P&IDs
v
Loop drawings
v
Calibration
v
Loop checks
v
Systematic troubleshooting

```

That sequence also reflects what ISA's technician framework is trying to develop: the ability to **calibrate, inspect, troubleshoot, commission, loop-check, and document actual control systems**, not simply recognize instrumentation vocabulary.

And I'd give **troubleshooting disproportionately more repetitions**. Practitioner feedback consistently calls fault-finding one of the major distinctions between someone who can perform routine instrument work and someone who really understands the system.

The goal is eventually to look at something like:

PT-101 reads correctly locally, PLC shows 0%, loop is 4-20 mA, 24 VDC supply is healthy

...and have your brain automatically start building the signal path and asking **where the known-good signal stops**, rather than reaching for a memorized list of possible bad parts. That's the kind of instinct these prompts should be drilling.

Focused Drill: Blank-Page Terminal and Loop Wiring Reconstruction

Inputs for This Specific Template

For this drill, I would **keep [device] and [loop] narrower than the overall I&C fundamentals list**.

ISA-5.4 treats loop diagrams as representations of the actual physical interconnections, energy sources, terminations, and devices in an individual instrument loop. ISA training material also stresses that loop checks catch wrong terminations, broken conductors, grounding problems, and configuration/scaling errors.

Practitioner feedback points in the same direction: being able to look at a print, understand the intended circuit, and trace power/signal through it is repeatedly cited as a core technician skill.

[device] - Best Inputs

I would drill these approximately in this order:

Core - repeat these heavily

- 2-wire pressure transmitter
- 2-wire differential-pressure transmitter
- 2-wire level transmitter
- 2-wire flow transmitter
- 2-wire temperature transmitter
- head-mounted temperature transmitter
- RTD connected to a 2-wire temperature transmitter
- thermocouple connected to a 2-wire temperature transmitter

Then add

- 2-wire radar level transmitter
- 2-wire guided-wave radar level transmitter
- 2-wire ultrasonic level transmitter
- 2-wire DP flow transmitter
- 2-wire hydrostatic level transmitter
- remote-mounted temperature transmitter
- 4-wire pressure transmitter
- 4-wire flow transmitter
- 4-wire level transmitter
- externally powered temperature transmitter

Later, for more complicated wiring

- HART pressure transmitter
- HART DP transmitter
- HART level transmitter
- HART temperature transmitter
- transmitter through an intrinsic-safety barrier
- transmitter through a galvanic isolator
- transmitter through a signal isolator

Best Starting [device] Bank

If you only want **10**, use:

2-wire pressure transmitter
 2-wire differential-pressure transmitter
 2-wire level transmitter
 2-wire flow transmitter
 2-wire temperature transmitter
 RTD with head-mounted temperature transmitter
 thermocouple with head-mounted temperature transmitter
 2-wire radar level transmitter
 HART pressure transmitter
 transmitter through an intrinsic-safety barrier

These force you to learn the reusable architecture rather than memorizing one specific instrument.

[loop] - Best Inputs

This bracket should describe the **actual topology**, not merely say "pressure loop."

Tier 1 - Fundamental 4-20 mA Wiring

Start here.

- external 24 VDC supply -> 2-wire transmitter -> passive PLC analog input
- external 24 VDC supply -> 2-wire transmitter -> passive DCS analog input
- active PLC analog input -> 2-wire transmitter
- active DCS analog input -> 2-wire transmitter
- 24 VDC supply -> field junction box -> transmitter -> junction box -> PLC analog input
- 24 VDC supply -> marshalling terminals -> transmitter -> PLC analog input

These are the repetitions I'd hammer first. A loop drawing's value is precisely that it shows field devices, wiring, junctions, terminations, energy sources, and control-system connections.

Tier 2 - Measurement-Specific Loops

- pressure transmitter -> PLC analog input
- differential-pressure transmitter -> PLC analog input
- DP flow transmitter -> PLC analog input
- hydrostatic level transmitter -> PLC analog input
- radar level transmitter -> PLC analog input
- guided-wave radar transmitter -> PLC analog input
- flow transmitter -> PLC analog input
- temperature transmitter -> PLC analog input
- RTD -> temperature transmitter -> PLC analog input
- thermocouple -> temperature transmitter -> PLC analog input

The last two are particularly useful because they force you to distinguish the **sensor-side wiring** from the **transmitter's 4-20 mA output loop**.

Tier 3 - Intermediate Field Wiring

Once the simple circuit is automatic:

- 24 VDC supply -> transmitter -> junction box -> marshalling cabinet -> PLC analog input
- 24 VDC supply -> transmitter -> terminal strip -> PLC analog input
- transmitter -> field junction box -> control-room terminal strip -> analog input
- transmitter -> remote I/O analog input
- transmitter -> signal isolator -> PLC analog input
- transmitter -> galvanic isolator -> PLC analog input
- transmitter -> intrinsic-safety barrier -> PLC analog input

These are good because real loop diagrams commonly contain the intermediate terminations that disappear from simplified textbook drawings. ISA notes that loop diagrams are specifically used to document those installation details and later support troubleshooting.

Tier 4 - HART

- HART pressure transmitter -> HART-capable analog input
- HART DP transmitter -> HART-capable analog input
- HART temperature transmitter -> HART-capable analog input
- HART transmitter -> junction box -> HART-capable PLC analog input
- HART transmitter -> intrinsic-safety barrier -> HART-capable analog input

The point here is still **4-20 mA wiring first**. HART adds digital communication on top of the analog loop; it does not replace understanding the basic current path.

Tier 5 - Fault-Reconstruction Loops

These are especially valuable once normal wiring becomes comfortable.

Use [loop] as:

- 4-20 mA loop with reversed transmitter polarity
- 4-20 mA loop with open return conductor
- 4-20 mA loop with missing 24 VDC supply
- 4-20 mA loop landed on the wrong analog input channel
- 4-20 mA loop with reversed analog-input polarity
- 4-20 mA loop with incorrect common termination
- 4-20 mA loop with shield grounded at both ends
- 4-20 mA loop with shield improperly bonded to a signal conductor
- 4-20 mA loop with an open terminal
- 4-20 mA loop with a bypassed intrinsic-safety barrier

ISA calibration guidance specifically identifies wrong wiring points, broken conductors, and ground-loop issues as problems that loop checking is intended to reveal.

Best [device] + [loop] Combinations

These are the combinations I would actually feed your prompt:

[device][loop]	
2-wire pressure transmitter	external 24 VDC supply -> transmitter -> passive PLC analog input
2-wire DP transmitter	active PLC analog input -> transmitter
2-wire level transmitter	24 VDC supply -> junction box -> transmitter -> PLC analog input
2-wire flow transmitter	transmitter -> field junction box -> marshalling terminals -> PLC analog input
2-wire temperature transmitter	external 24 VDC supply -> transmitter -> passive PLC analog input
RTD with head-mounted temperature transmitter	RTD -> temperature transmitter -> 4-20 mA PLC analog input
thermocouple with temperature transmitter	thermocouple -> temperature transmitter -> 4-20 mA PLC analog input
HART pressure transmitter	HART transmitter -> HART-capable analog input
2-wire pressure transmitter	transmitter -> galvanic isolator -> PLC analog input
2-wire pressure transmitter	transmitter -> intrinsic-safety barrier -> PLC analog input

Especially good first drill

[device] = 2-wire pressure transmitter

[loop] = external 24 VDC power supply -> field transmitter ->
passive PLC analog input -> DC return

Then change **only one variable at a time**:

2-wire pressure transmitter
v
DP transmitter
v
temperature transmitter
v
level transmitter

Then change the topology:

External supply + passive AI
v
Active AI
v
Junction box
v
Marshalling terminals
v
Isolator
v
Intrinsic-safety barrier

That progression gives you the reusable mental model experienced techs rely on: **identify the source -> trace the energy/signal path -> follow every termination -> establish where the expected current stops.**

One Important Correction to the Template

I would **not** put these directly into [device] with the prompt exactly as written:

- bare RTD
- bare thermocouple
- 3-wire RTD
- 4-wire RTD

They do **not** use the same 24 VDC -> field transmitter -> 4-20 mA -> analog input topology. A direct RTD terminates at an RTD/resistance input, while a thermocouple produces a millivolt signal and terminates at a TC-compatible input. Treating them as ordinary 4-20 mA loops would teach the wrong mental model.

Use them here instead:

[device] = RTD with head-mounted temperature transmitter

or

[device] = thermocouple with head-mounted temperature transmitter

Then later build a separate **Temperature Element Wiring Reconstruction** drill specifically for 2-, 3-, and 4-wire RTDs and thermocouple polarity/cold-junction wiring.